

Finite Weil Fibres for a Rauzy–AGY Transfer Operator: Fredholm Determinants, Quadratic Traces, and Chronology Collapse

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Abstract

For every fixed odd prime, we construct an ordinary Fredholm determinant by twisting a source-locked genus-two Rauzy–AGY transfer operator with the genuine finite Weil representation of $\mathrm{Sp}(4, \mathbb{F}_p)$. The fibre has dimension p^2 , so the scalar Bergman trace-class theorem survives without regularization, and a periodic word acquires the exact arithmetic factor $\Theta_p(g) = \mathrm{Tr} \rho_p(g)$. Thomas’s character formula makes this factor reproducible: at primes not dividing $\det(g - I)$, it is the Legendre symbol $\left(\frac{\det(g-I)}{p}\right)$. For a frozen noncyclic three-return chronology, the complete finite-fibre polynomials $\det(I - T\rho_p(g))$ differ at $p = 3, 5, 7$; among 576 exact comparisons with odd $p \leq 97$ and $1 \leq r \leq 24$, 328 power characters differ. Two exact obstructions show why this signal does not yet define a Hilbert–Pólya structure. At $p = 43$, two matrices with different characteristic polynomials have order 925 and equal Weil characters at every power, hence equal degree-1849 finite-fibre polynomials. More strongly, two distinct primitive symbolic cycles are conjugate in $\mathrm{Sp}(J_{24}, \mathbb{Z})$, so every class-function fibre collapses their entire repetition towers at every prime. Finally, 150 source-locked branches have 150 distinct quadratic signatures over the odd primes below 100. Finite Weil fibres therefore repair the ordinary-determinant obstruction and add genuine arithmetic phases, but the modulus remains external and the observed arithmetic fragments by orbit.

1 Introduction

A finite Weil fibre opens a determinant door that the infinite oscillator fibre leaves closed. For the genus-two Rauzy–AGY cocycle considered here, the oscillator representation acts on an infinite-dimensional Hilbert space and produces a noncompact unsmoothed transfer operator. Reducing the same chronological cocycle modulo an odd prime replaces that fibre by a canonical unitary representation of dimension p^2 . The resulting twisted operator is trace class on the existing common Bergman domain.

Existence of a determinant is only the first gate. A dynamical candidate for Hilbert–Pólya must also retain orbit chronology, produce arithmetic structure without reading prime or zero tables, and resist collapse under natural equivalences. Finite Weil characters are plausible because they are exact Gauss-sum invariants of the pathwise symplectic product. They are also class functions, which makes a sharp negative outcome possible: symbolic histories can remain distinct while their cocycles become symplectically conjugate.

This paper evaluates both sides of that tradeoff. We freeze the seven-state Rauzy source, the fixed-fibre symplectic trivialization, the raw AGY roof, and the common complex domain from the preceding scalar construction. For each odd prime p , we use the genuine Weil representation ρ_p of $\mathrm{Sp}(4, \mathbb{F}_p)$, not a projective representation and not a truncation of the real oscillator. We evaluate a

character only after forming the complete later-on-the-left path product. This order convention is essential: Weil characters are not multiplicative.

Our contributions are concrete and falsifiable.

- (i) We prove that the finite-Weil twist is trace class for every fixed odd prime and derive the exact word atom

$$\Theta_p(g_w) \frac{\lambda_w^{-(s+1)}}{\chi'_w(\lambda_w)}.$$

Its Fredholm determinant is jointly holomorphic on the same source half-plane as the scalar determinant.

- (ii) We implement Thomas’s full character formula over $\mathbb{F}_p$, including singular fixed-vector loci, and reconstruct the finite-fibre polynomial $\det(I - T\rho_p(g))$ from all p^2 power traces.
- (iii) We certify genuine chronology sensitivity. The three-return forward and noncyclic reverse polynomials differ at $p = 3, 5, 7$, while the required two-return AB/BA null control remains equal.
- (iv) We prove two limits of the method. The same three-return pair has an equal complete finite-fibre polynomial at $p = 43$, and an explicit integral symplectic conjugacy makes every class-function fibre blind to a distinct pair of primitive symbolic cycles for all primes and repetitions.

The arithmetic census gives the broader verdict. At good primes a word trace reduces to the quadratic character associated with the squarefree part of $D_w = \det(g_w - I)$. In a finite source-locked scan, all 150 branches have different Legendre signatures across the odd primes at most 97. The data therefore support an orbit-dependent family of quadratic fields rather than a small common conductor. No prime product, adelic measure, or limit in p is introduced to hide that fragmentation.

The paper proceeds from the source and order conventions to the finite Weil character, the Fredholm theorem, and the exact chronology experiments. The integral conjugacy result then isolates a theorem-level obstruction that no larger character table can remove. We close by stating the precise Route-A boundary: fixed-prime determinants are natural and rigorous, but a global Hilbert–Pólya object remains absent.

2 Related structures

Rauzy–Veech and AGY dynamics. Rauzy induction records interval-exchange dynamics by a directed symbolic system whose integral matrices multiply in chronological order. The accelerated model of Avila et al. [1] supplies a countable Markov return map, a logarithmic roof, and the exponential-tail estimates needed for transfer operators. Our source is a frozen four-letter, genus-two instance of that framework. We do not alter its clock, average its transition matrices, or identify paths merely because they share a characteristic polynomial. The new question is what arithmetic information survives when the same integral symplectic cocycle is reduced modulo primes.

Holomorphic transfer operators. Compactly contained holomorphic map-weight systems admit strong eigenvalue and trace-ideal bounds on Bergman-type spaces [2, 3]. The scalar companion construction provides exactly such a common domain for the countable AGY branches and proves locally uniform trace-norm summability. The present paper does not rederive that complex-cone theorem. Instead, it asks whether an arithmetic vector fibre preserves the scalar result. Finite dimension is decisive: tensoring a branch with a p^2 -dimensional unitary multiplies its trace norm by p^2 , whereas the unsmoothed real oscillator has infinite multiplicity.

Weil representations over finite and real fields. The oscillator or metaplectic representation over $\mathbb{R}$ is naturally tied to symplectic phase space [4]. Over a finite field of odd characteristic, the canonical intertwining construction yields an honest linear Weil representation of the symplectic group [5]. The genus-two fibre used here is the full p^2 -dimensional representation. In particular, we do not add edgewise metaplectic signs, quotient by the central element $-I$, or silently retain only an even or odd parity constituent.

Exact characters and dynamical weights. Thomas [6] expresses the finite Weil character through the fixed space of g and the discriminant of a bilinear form on $V/\ker(g - I)$. That theorem makes singular primes computable without a dense p^2 -by- p^2 matrix. Our use differs from a pointwise character Euler product: the character is evaluated on a genuine chronological product, and repetitions use $\Theta_p(g^r)$, not $\Theta_p(g)^r$. We also reconstruct the complete characteristic polynomial of the fibre operator by Newton identities.

Relation to Hilbert–Pólya searches. A trace-class dynamical determinant and a natural unitary fibre address only two structural requirements of a Hilbert–Pólya programme. A successful global construction would still need an intrinsic prime organization, a controlled analytic continuation, the appropriate functional equation and gamma factor, and a bridge to a self-adjoint spectral problem. The present work treats the absence of those features as a result, not as a normalization detail. Its main negative controls quantify information lost by every class-function fibre before any comparison with Riemann zeros is attempted.

3 The source-locked Rauzy–AGY system

3.1 Fixed symplectic fibre

The source is the four-letter Rauzy class with seven states and fourteen labeled arrows. Each edge $e : \pi \rightarrow \pi'$ has an integral chronological matrix B_e . State-dependent alternating forms are trivialized by integral frames S_π , giving

$$g_e = S_{\pi'}^{-1} B_e S_\pi, \quad g_e^T J_0 g_e = J_0,$$

where

$$J_0 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 1 & 0 \end{pmatrix}.$$

For a forward path $w = e_1 \cdots e_n$,

$$g_w = g_{e_n} \cdots g_{e_1} = S_{\pi_n}^{-1} B_w S_{\pi_0}. \tag{1}$$

All intermediate frames telescope. The integral Darboux basis

$$T = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \det T = -1,$$

satisfies

$$T^T J_0 T = J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}.$$

Because T is unimodular, this identification is valid modulo every prime.

3.2 Scalar holomorphic operator

Let Ω be the common bounded complex projective domain from the scalar source theorem, and let Γ be the intrinsic countable AGY return alphabet. A branch γ has inverse map h_γ , positive normalizer q_γ , and raw weight

$$w_{s,\gamma}(z) = \exp(-(s+4) \operatorname{Log} q_\gamma(z)).$$

The principal logarithm is common to all branches. On $\mathcal{H}_0 = A^2(\Omega)$, the scalar operator

$$(L_s f)(z) = \sum_{\gamma \in \Gamma} w_{s,\gamma}(z) f(h_\gamma z)$$

is trace class for $s \in \{s \in \mathbb{C} : \operatorname{Re} s > -\sigma_0\}$, and the branch series converges locally uniformly in trace norm. This statement follows from the common compact containment and the holomorphic transfer estimates of Bandtlow and Jenkinson [2, 3].

For a periodic operator word with positive matrix A_w , Perron root λ_w , and characteristic polynomial $\chi_w(t) = \det(tI - A_w)$, the scalar fixed-point trace is

$$\operatorname{Tr} K_{s,w} = \frac{\lambda_w^{-(s+1)}}{\chi'_w(\lambda_w)}. \quad (2)$$

3.3 Two orders, one chronological product

The forward and operator-factor labels must not be conflated. If $(\beta_1, \dots, \beta_n)$ is the forward Rauzy order, then

$$g_{\text{fwd}} = g_{\beta_n} \cdots g_{\beta_1}.$$

The associated transfer-operator factors occur as $(\beta_n, \dots, \beta_1)$. With column fibres and left matrix action, their product is therefore

$$\rho_{p,J_0}(g_{\beta_n}) \cdots \rho_{p,J_0}(g_{\beta_1}) = \rho_{p,J_0}(g_{\text{fwd}}).$$

This contravariant bookkeeping preserves, rather than reverses, the forward cocycle chronology.

We retain two permanent controls. A two-return AB/BA pair has distinct matrices but equal characters in every representation by trace cyclicity. A three-return forward/noncyclic-reverse pair has integer characteristic polynomials whose nonzero coefficient differences have greatest common divisor 64, so those base characteristic polynomials remain different modulo every odd prime.

4 The finite Weil character

4.1 Convention and honest representation

Fix an odd prime p . In standard coordinates let

$$V = \mathbb{F}_p^2 \oplus (\mathbb{F}_p^2)^\vee, \quad J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \quad \psi_p(a) = e^{2\pi i \tilde{a}/p}.$$

For $v = (x, \xi)$, use the Weyl operator

$$[\pi(x, \xi)f](t) = \psi_p(\xi^T(t + x/2)) f(t + x) \quad \text{on } \mathbb{C}^{\mathbb{F}_p^2}.$$

The canonical intertwining construction gives an honest representation

$$\rho_p : \mathrm{Sp}(4, \mathbb{F}_p) \longrightarrow U(p^2)$$

satisfying

$$\rho_p(g)\pi(v)\rho_p(g)^{-1} = \pi(gv).$$

This is a representation of $\mathrm{Sp}(4, \mathbb{F}_p)$, not a choice of signs on a double cover [5]. The central element $-I$ acts by parity and cannot be discarded by passing to the projective symplectic group.

More generally, let J_* be any frozen unimodular alternating form and choose an integral Darboux matrix T_* satisfying $T_*^T J_* T_* = J$. We pull the representation back by

$$\rho_{p,J_*}(g) := \rho_p(T_*^{-1}gT_*), \quad g \in \mathrm{Sp}(J_*, \mathbb{F}_p). \quad (3)$$

Changing the Darboux basis conjugates all fibre operators and leaves their characters and fibre polynomials unchanged. For the C25/C26 cocycle we take $J_* = J_0$ and the fixed matrix T from section 3; the C24 controls use their displayed form J_{24} . Whenever the source frame is clear, we abbreviate

$$\Theta_p(g) := \mathrm{Tr} \rho_{p,J_*}(g).$$

4.2 Thomas's quotient discriminant

Let $g \in \mathrm{Sp}(J_*, \mathbb{F}_p)$, set $H_g = g - I$, and write

$$k_g = \dim \ker H_g.$$

On $V/\ker H_g$, define

$$\sigma_g(\bar{v}, \bar{w}) = \langle H_g v, w \rangle_{J_*}.$$

If the columns of C represent a basis of the quotient, then

$$d_g = \det(C^T H_g^T J_* C) \in \mathbb{F}_p^\times$$

is defined up to a square. Let η_p be the Legendre character and

$$G_p = \sum_{x \in \mathbb{F}_p} \psi_p(x^2/2), \quad G_p^2 = \eta_p(-1)p.$$

Theorem 4.1 (Thomas character formula 6). *With the preceding convention,*

$$\Theta_p(g) := \mathrm{Tr} \rho_{p,J_*}(g) = p^{k_g/2} \gamma_p(1)^{4-k_g} \eta_p(d_g), \quad \gamma_p(1) = p^{-1/2} G_p.$$

Equivalently, every character lies in the exact quadratic span $\mathbb{Z} + \mathbb{Z}G_p$ and is given by

$$\Theta_p(g) = \begin{cases} \eta_p(d_g), & k_g = 0, \\ \eta_p(d_g)\eta_p(-1)G_p, & k_g = 1, \\ \eta_p(d_g)\eta_p(-1)p, & k_g = 2, \\ \eta_p(d_g)pG_p, & k_g = 3, \\ p^2, & k_g = 4. \end{cases} \quad (4)$$

The table yields three strong implementation identities:

$$\Theta_p(I) = p^2, \quad |\Theta_p(g)|^2 = p^{k_g} = |\ker(g - I)|, \quad \Theta_p(g^{-1}) = \overline{\Theta_p(g)}. \quad (5)$$

In particular, the full Weil character never vanishes. Character values are conjugacy invariant but not multiplicative. An orbit weight must therefore be $\Theta_p(g_w)$, never a product of edge or branch characters.

4.3 Good primes and quadratic fields

The character simplifies strongly away from the fixed-vector locus.

Proposition 4.2 (Good-prime quadratic law). *Let J_* be a unimodular integral alternating form, let $g \in \mathrm{Sp}(J_*, \mathbb{Z})$, and let $D_g = \det(g - I) \neq 0$. For every odd prime $p \nmid D_g$,*

$$\Theta_p(g \bmod p) = \left(\frac{D_g}{p} \right).$$

The same statement holds for g^r whenever $p \nmid \det(g^r - I)$.

Proof. The condition $p \nmid D_g$ gives $k_g = 0$. In full quotient coordinates, the determinant of σ_g differs from $\det(g - I)$ by the determinant of the unimodular alternating form, which is a square. The $k = 0$ line of equation (4) gives the claim. $\square$

Thus a fixed orbit determines a genuine quadratic character through the squarefree part of D_g . The primes dividing D_g are precisely where the fixed-space dimension and quotient discriminant can add Gauss-sum data. The simplicity is also a warning: at every good prime a p^2 -dimensional character has collapsed to one bit, $+1$ or -1 .

4.4 Complete finite-fibre polynomial

The first character alone does not encode repetitions. Define

$$P_{p,g}(T) = \det(I - T\rho_{p,J_*}(g)).$$

Then

$$-\log P_{p,g}(T) = \sum_{r \geq 1} \frac{\Theta_p(g^r)}{r} T^r. \quad (6)$$

Because the degree is p^2 , the first p^2 traces determine the polynomial. Writing $P_{p,g}(T) = \sum_{m=0}^{p^2} c_m T^m$, Newton's recursion is

$$c_0 = 1, \quad c_m = -\frac{1}{m} \sum_{j=1}^m c_{m-j} \Theta_p(g^j). \quad (7)$$

All computations can be performed exactly in $\mathbb{Q}(G_p)$. The determinant character of ρ_p is trivial in genus two, so $c_{p^2} = -1$; unitarity also gives

$$c_m = -\overline{c_{p^2-m}}.$$

These identities are used as release tests. The polynomial $P_{p,g}$ is the complete local invariant of the finite fibre. It is not, by itself, the full AGY orbit factor, which also contains the scalar Perron and stability terms in equation (2).

5 A fixed-prime Fredholm determinant

For a fixed odd prime p , put

$$\mathcal{H}_p = A^2(\Omega) \otimes \mathbb{C}^{p^2}.$$

The branch action uses column fibres and left multiplication:

$$(\mathcal{L}_{s,p}F)(z) = \sum_{\gamma \in \Gamma} w_{s,\gamma}(z) \rho_{p,J_0}(g_\gamma \bmod p) F(h_\gamma z). \quad (8)$$

Theorem 5.1 (Finite-Weil Fredholm theorem). *For every fixed odd prime p and every $s \in \{s \in \mathbb{C} : \operatorname{Re} s > -\sigma_0\}$, the operator $\mathcal{L}_{s,p}$ is trace class on $\mathcal{H}_p$. Its singular values remain in the same exponential class as the scalar operator, with finite multiplicity p^2 . Moreover,*

$$D_p(s, u) = \det_{\mathcal{H}_p}(I - u\mathcal{L}_{s,p})$$

is jointly holomorphic on $\{s \in \mathbb{C} : \operatorname{Re} s > -\sigma_0\} \times \mathbb{C}$.

Proof. Let $K_{s,\gamma}$ be the scalar weighted-composition branch. Since $\rho_{p,J_0}(g_\gamma)$ is unitary on a p^2 -dimensional space,

$$\|K_{s,\gamma} \otimes \rho_{p,J_0}(g_\gamma)\|_1 = p^2 \|K_{s,\gamma}\|_1.$$

The scalar branch series is locally uniformly summable in trace norm on compact subsets of $\{s \in \mathbb{C} : \operatorname{Re} s > -\sigma_0\}$. Multiplication by the fixed number p^2 preserves that property, so the twisted branch series converges in trace norm. More explicitly, the whole matrix-weighted sum factors through $R \otimes I_{p^2}$, where R is the common compact Bergman restriction from C26, followed by a bounded matrix-weighted branch sum. The singular values of this common restriction are therefore repeated only with finite multiplicity p^2 . Trace-norm holomorphy and the standard trace-class determinant theorem give joint holomorphy in (s, u) . $\square$

Theorem 5.2 (Chronological word atom). *Let $(\beta_1, \dots, \beta_n)$ be a forward periodic Rauzy word, let*

$$g_w = g_{\beta_n} \cdots g_{\beta_1},$$

and let λ_w and χ_w be the scalar Perron data from equation (2). The corresponding twisted operator word has trace

$$\boxed{\operatorname{Tr} K_{s,p,w} = \Theta_p(g_w) \frac{\lambda_w^{-(s+1)}}{\chi'_w(\lambda_w)}}.$$

For its r -fold repetition, the fibre factor is $\Theta_p(g_w^r)$, not $\Theta_p(g_w)^r$.

Proof. The operator-factor order is $(\beta_n, \dots, \beta_1)$. The representation property gives the fibre product $\rho_{p,J_0}(g_w)$, while the scalar fixed-point trace is equation (2). The trace of a tensor product is the product of the two traces. Repeating the operator word replaces $\rho_{p,J_0}(g_w)$ by $\rho_{p,J_0}(g_w^r)$. $\square$

Remark 5.3 (What the theorem does not assemble). *The family $\{D_p\}_{p \text{ odd}}$ is not a single determinant. The modulus p is an external parameter, and the fibre dimensions grow with p . Neither an unregularized product over primes nor a direct sum over all finite fibres is justified by theorem 5.1. A global construction would require a measure, trace, and convergence law derived from the dynamics rather than appended after the computation.*

6 Exact finite-field results

6.1 Reproducible protocol

The producer reads cryptographically frozen C24, C25, and C26 certificates. It forms each integral chronological product before reducing modulo p , computes a quotient basis for $V/\ker(g - I)$, evaluates equation (4), and stores the result as an integer pair in the basis $(1, G_p)$. No floating-point character value enters a decision. An independent checker does not import the producer; it uses a separate tuple-matrix implementation, Bareiss determinants, and independent modular elimination.

Mutation tests reject forward/reverse substitution, B^T in place of the homology cocycle, chronological averaging, products of branch characters, $\Theta(g)^r$ in place of $\Theta(g^r)$, inclusion of $p = 2$, and promotion of a finite window to an all-power theorem.

6.2 Small-prime chronology and complete fibre polynomials

Let g_{fwd} and g_{rev} be the frozen C26 three-return forward and noncyclic reverse matrices. The exact first characters are:

Table 1: The three-return chronology is distinguished at all three small primes. The basis satisfies $G_p^2 = \eta_p(-1)p$.

p	fibre dimension	$\Theta_p(g_{\text{fwd}})$	$\Theta_p(g_{\text{rev}})$
3	9	1	G_3
5	25	-1	1
7	49	-1	$-G_7$

Newton reconstruction uses all p^2 power traces. The two fibre polynomials differ for each of $p = 3, 5, 7$; their coefficient hashes are listed in table 2. The leading coefficient is -1 , and every polynomial passes conjugate reciprocity.

Table 2: SHA-256 prefixes of exact ascending coefficient arrays in the $(1, G_p)$ basis. Different hashes certify different full fibre polynomials; the JSON artifact contains every coefficient.

p	forward	noncyclic reverse
3	77c8f89433b015e7	1565b7a342d05194
5	26a11d78dd2e464a	05d24ca6db4eebc2
7	087b3cdec27e34e6	1d86d5bce7c3caa7

The two-return AB/BA matrices remain a null control. They differ over $\mathbb{Z}$, but $\text{Tr } \rho_{p, J_0}(AB) = \text{Tr } \rho_{p, J_0}(BA)$ for every p . A program that reports sensitivity on this pair has manufactured a phase or order error.

6.3 Power scan and a complete $p = 43$ fibre collision

For every odd prime $p \leq 97$ and $1 \leq r \leq 24$, we compare $\Theta_p(g_{\text{fwd}}^r)$ and $\Theta_p(g_{\text{rev}}^r)$. Among 576 comparisons, 328 differ and 248 agree. The patterns are structured rather than monotone; table 3 compresses the exact census.

The last row cannot be interpreted from a cutoff. Extending the exact scan finds the first differences at $r = 41$ for $p = 83$ and $r = 30$ for $p = 89$. Prime 43 behaves differently.

Table 3: Power-character patterns in the range $1 \leq r \leq 24$.

Distinguished powers	primes
all $1, \dots, 24$	5, 7, 31, 53, 61, 73, 97
all except 18	3
odd powers	11, 67, 79
even powers	13, 19, 29, 41, 59, 71
9 and 18 only	17
24 only	23
none in the window	43, 83, 89

Proposition 6.1 (Complete $p = 43$ finite-fibre collision). *Modulo 43, both g_{fwd} and g_{rev} have order 925. Their Weil characters agree for every $1 \leq r \leq 925$. Consequently*

$$P_{43, g_{\text{fwd}}}(T) = P_{43, g_{\text{rev}}}(T),$$

even though the characteristic polynomials of the two base symplectic matrices are different modulo 43.

Proof. Exact modular multiplication returns both powers to the identity first at 925. Thomas's formula gives equal character pairs at every preceding power. The two sequences are therefore equal for all $r \geq 1$ by periodicity, and equation (6) identifies the fibre polynomials. Direct reduction gives base characteristic coefficients

$$(1, 33, 9, 33, 1) \quad \text{and} \quad (1, 11, 13, 11, 1) \pmod{43},$$

so the base matrices are spectrally distinct. □

The scalar Perron atoms remain different in proposition 6.1; hence the full twisted AGY orbit contributions need not collide. The proposition proves only, and exactly, that a complete finite Weil fibre is not a separating invariant.

6.4 Singular-prime refinement

The C24 controls P014 and P016 share

$$\chi(t) = t^4 - 9t^3 + 19t^2 - 9t + 1, \quad \det(I - g) = 3.$$

At the singular prime 3, both fixed spaces have dimension one, but their quotient discriminants have opposite Legendre signs. In the frozen normalization,

$$\Theta_3(g_{014}) = -G_3, \quad \Theta_3(g_{016}) = +G_3.$$

This is information beyond the common characteristic polynomial. Away from 3, proposition 4.2 forces both traces to the same Legendre symbol.

6.5 Arithmetic fragmentation in 150 branches

We enumerate every first-return bridge to the AGY base of length at most 12, with no internal base visit. For each bridge v , the tested branch is

$$\gamma_* v \gamma_*, \quad \gamma_* = t^{64}(tbttbtbb)^8.$$

There are 150 such bridges. The exact scan finds

150 distinct $D_v = \det(I - g_v)$,
 150 distinct characteristic polynomials,
 150 distinct Legendre signatures

over the 24 odd primes from 3 through 97. Four released reference orbits also have four different squarefree kernels; the forward and reverse three-return discriminants have disjoint-looking large factor patterns.

This census is finite and does not prove all-length conductor fragmentation. It does falsify the first, strongest version of the arithmetic hypothesis: the tested branches do not concentrate into a small set of quadratic characters. Extending the bridge cutoff would add data but would not repair the missing mechanism that should force a common conductor.

7 Integral conjugacy collapse

The finite-field results show that one Weil fibre is not separating. The underlying Rauzy graph contains a stronger obstruction that no collection of class characters can overcome.

Consider the two primitive C24 control cycles P076 and P082. After a cyclic move to their common central state, their first-return orders are

$$[bbb, tbbt, tbbtt] \quad \text{and} \quad [bbb, tbbtt, tbbt].$$

They use the same three labeled return branches in different noncyclic order. The canonical cycles are distinct and their chronological matrices remain different modulo every prime. In the C24 certificate these are the `base_trivialized_symplectic_matrix` entries in the fixed frame rooted at state 2:

$$A = \begin{pmatrix} 2 & 1 & 0 & 3 \\ 2 & 3 & 0 & 4 \\ 3 & 5 & 2 & 6 \\ 2 & 3 & 1 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 1 & 4 & 0 & 4 \\ 1 & 6 & 2 & 5 \\ 1 & 4 & 1 & 4 \end{pmatrix}.$$

Both preserve the frozen C24 form

$$J_{24} = \begin{pmatrix} 0 & -1 & 1 & -1 \\ 1 & 0 & -1 & 1 \\ -1 & 1 & 0 & -1 \\ 1 & -1 & 1 & 0 \end{pmatrix}$$

and have characteristic polynomial

$$t^4 - 11t^3 + 18t^2 - 11t + 1.$$

Theorem 7.1 (Integral symplectic chronology collapse). *The matrices A and B are conjugate in $\mathrm{Sp}(J_{24}, \mathbb{Z})$. Explicitly,*

$$X = \begin{pmatrix} 0 & 0 & -1 & 1 \\ -1 & 0 & 0 & -1 \\ 0 & -2 & 0 & -1 \\ 0 & -1 & 0 & -1 \end{pmatrix}$$

satisfies

$$\det X = 1, \quad X^T J_{24} X = J_{24}, \quad BX = XA.$$

Consequently, for every prime p , every integer r , and every class function φ_p on $\mathrm{Sp}(J_{24}, \mathbb{F}_p)$,

$$\varphi_p(A^r) = \varphi_p(B^r).$$

In particular,

$$\Theta_p(A^r) = \Theta_p(B^r) \quad \text{for every odd } p \text{ and every } r \geq 1.$$

Proof. The three displayed integer identities are verified by direct multiplication. Because $\det X = 1$, reduction modulo any prime leaves X invertible and symplectic. Thus $B^r = X A^r X^{-1}$ over every $\mathbb{F}_p$. A class function is constant on the resulting conjugacy class. $\square$

This collapse is stronger than equality of the first Weil character. It identifies the full power-character towers, every parity constituent, and every linear combination of irreducible characters. Since A and B also have the same homological characteristic polynomial, their homological Perron and characteristic data agree throughout the repetition tower. Within the C24 control census, neither those data nor any class-function fibre recovers the symbolic order.

The scope matters. P076 and P082 are controls from the underlying finite Rauzy graph; they are not promoted to branches of the source-locked C26 positive-prefix AGY operator. The theorem therefore establishes an information-theoretic obstruction for class-function quantizations of the underlying cocycle, not an equality between two terms already identified in the C26 Fredholm series. Finding a control inside the accelerated C26 alphabet would strengthen the application but would not change the mechanism.

Obstruction 7.2 (Class functions cannot be chronology-complete). *Any fibre weight that depends on a chronological cocycle only through its symplectic conjugacy class loses some labeled symbolic chronology in this Rauzy class. Enlarging the finite character table or taking all prime reductions cannot distinguish P076 from P082.*

To preserve the missing information, a future construction would have to retain a based or path-groupoid observable rather than only a conjugacy-class trace. Such an observable would require a new determinant theorem: ordinary periodic-orbit traces are cyclic and naturally descend to conjugacy data.

8 Route-A implications and claim boundary

The fixed-prime construction is mathematically stronger than the failed infinite-oscillator realization: the target operator itself now has an ordinary determinant. It is nevertheless not a global Riemann dynamical determinant. Table 4 separates the four evaluation layers.

The resulting tuple is

$$(A1_WEAK, A2_ANALYTIC_DETERMINANT, \\ A3_PARTIAL_ANALYTIC_STRUCTURE, A4_NATURAL_QUANTIZATION)$$

with overall status *Route-A exploratory*. Route B is not authorized.

Table 4: Conservative Route-A assessment. “Analytic determinant” refers to the fixed-prime target, not to a divisor match with the completed zeta function.

Layer	verdict	reason
A1	weak	Primitive Rauzy words are intrinsic and exact, but no prime-like clock or amplitude law is known; the quadratic conductors fragment in the first 150-branch census.
A2	analytic determinant	For each fixed odd p , $D_p(s, u)$ is an ordinary trace-class Fredholm determinant. No zero-divisor comparison has been attempted or licensed.
A3	partial analytic structure	The determinant is jointly holomorphic on $\{s \in \mathbb{C} : \text{Re } s > -\sigma_0\} \times \mathbb{C}$. No continuation to the full plane, functional equation, gamma factor, trivial-zero mechanism, or counting law is present.
A4	natural quantization	The finite Weil fibre is canonically induced by the exact symplectic cocycle and preserves the AGY clock. The prime modulus is external, and no self-adjoint global operator has been derived.

Why the arithmetic is real. No prime table is used to choose a path, clock, or parameter. Once a word is fixed, proposition 4.2 produces a bona fide quadratic character, and singular primes add exact Gauss phases. These weights are intrinsic to the finite reduction of the cocycle.

Why the arithmetic is insufficient. The good-prime character depends only on D_w up to square class. The observed words generate different square classes rather than a shared conductor. The complete $p = 43$ fibre collision and theorem 7.1 show that even all repetitions can lose order information. More computation at neighboring primes would refine this description but would not create a global assembly law.

The next large gate. A continuation of this route is justified only if the dynamics itself selects an adelic measure or trace for the family $\{\rho_p\}$, proves the convergence of the resulting determinant, and explains how orbit-dependent quadratic characters combine without changing the AGY clock. Multiplying the D_p by hand, taking an uncontrolled direct sum, or fitting prime weights would violate the source lock. If no intrinsic assembly appears, the appropriate next move is to change the dynamical form rather than extend the finite scan.

No statement in this paper matches Riemann zeros, proves a functional equation, constructs a self-adjoint operator with the zeta ordinates, or advances the Riemann hypothesis. The contribution is a fixed-prime determinant theorem together with exact sensitivity and collapse mechanisms.

9 Conclusion

Finite Weil fibres solve the immediate operator-theoretic problem left by the infinite oscillator: for each fixed odd prime, the source-locked AGY transfer operator has a genuine trace-class twist and an ordinary Fredholm determinant. Its periodic weights combine the scalar Perron atom with an

exact Legendre–Gauss character, and complete finite-fibre polynomials detect the frozen three-return chronology at $p = 3, 5, 7$.

The same experiment also identifies the boundary of that success. A complete period at $p = 43$ makes the finite fibre blind to the C26 order change even though the scalar matrices remain spectrally distinct. An explicit integral symplectic conjugacy produces a stronger all-prime, all-repetition collapse for two distinct C24 symbolic cycles. Meanwhile, 150 tested AGY branches yield 150 quadratic signatures rather than a common arithmetic conductor.

The fixed-prime family is therefore a natural arithmetic quantization, not a global Hilbert–Pólya operator. The next credible step must derive a canonical global prime assembly from the dynamics and prove its analytic control. Without that mechanism, extending the same small-prime census is a small door; the research programme should move to a different dynamical architecture.

Reproducibility. The release includes the exact producer, an independent implementation that does not import it, all coefficient arrays, mutation tests, source hashes, and a one-command replay. All mathematical decisions use integer, rational, or finite-field arithmetic.

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A Exact quotient computation

Let $H = g - I$ over $\mathbb{F}_p$, and choose pivot columns $i_1, \dots, i_r$ of H . The corresponding standard vectors $e_{i_1}, \dots, e_{i_r}$ project to a basis of $V/\ker H$: a linear combination lies in $\ker H$ exactly when the same combination of the independent columns of H vanishes. If $C = (e_{i_1}, \dots, e_{i_r})$, the quotient matrix is

$$Q_g = C^T H^T J C.$$

Thomas's form is nondegenerate, so $\det Q_g \neq 0$. Replacing C by another quotient basis sends Q_g to $R^T Q_g R$, multiplying the determinant by $(\det R)^2$. Its Legendre sign is therefore well defined.

The independent checker reimplements the pivot-column quotient construction with immutable tuple matrices and its own modular elimination. Independence here means a separate arithmetic engine and source replay, not a claim that a different quotient-basis theorem is being used.

B Arithmetic in the $(1, G_p)$ basis

Write an exact value as $a + bG_p$, where $G_p^2 = \eta_p(-1)p$. Multiplication and complex conjugation are

$$(a, b)(c, d) = (ac + bd\eta_p(-1)p, ad + bc),$$

$$\overline{(a, b)} = (a, \eta_p(-1)b).$$

Newton's recursion equation (7) may introduce rational pairs in this basis even though the resulting coefficients are algebraic integers; the quadratic basis need not be an integral basis. The release therefore stores reduced rational numerators and denominators rather than coercing each coefficient to an integer.

C Frozen chronology witnesses

The positive prefix is

$$\gamma_* = t^{64}(tbttbtbb)^8, \quad |\gamma_*| = 128.$$

Let $\gamma_2 = \gamma_* bttbtbb \gamma_*$ and $\gamma_3 = \gamma_* bbb \gamma_*$. The forward three-return product is

$$g_{\text{fwd}} = g_{\gamma_3} g_{\gamma_2} g_{\gamma_*},$$

whereas the tested noncyclic reverse is

$$g_{\text{rev}} = g_{\gamma_*} g_{\gamma_2} g_{\gamma_3}.$$

The two-return products $g_{\gamma_2} g_{\gamma_*}$ and $g_{\gamma_*} g_{\gamma_2}$ are the mandatory trace-cyclicity control. All matrices are reconstructed from labeled Rauzy edges; the release does not hand-enter the large three-return matrices.

D Complete-period logic

Suppose a, b in a finite group both satisfy $a^N = b^N = I$. If $\Theta(a^r) = \Theta(b^r)$ for $1 \leq r \leq N$, then equality holds for all integer r by reduction modulo N . At $p = 43$, exact modular multiplication gives $N = 925$ and no character difference in the full period. This converts a finite computation into an all-power theorem. By contrast, the absence of a difference for $p = 83$ and $p = 89$ through $r = 24$ has no such status; differences appear later at 41 and 30.

E Independent replay and release artifacts

From the project directory, the complete replay command is

```
bash code/run_c27.sh
```

It performs the following steps:

1. regenerate the exact certificate from the C24–C26 source locks;
2. run the independent checker;
3. execute chronology and scope mutation tests;
4. verify the regenerated artifacts against the frozen SHA-256 release manifest.

The primary certificate contains all local-polynomial coefficients, every power-pattern class for odd $p \leq 97$, the complete $p = 43$ period record, the integral symplectic conjugator, and all 150 arithmetic-scan rows. Finite enumerations carry explicit flags preventing their promotion to all-length theorems.

F Claim firewall

The release excludes $p = 2$, edgewise metaplectic signs over finite fields, branchwise products of characters, absolute-value replacements, parity-sector substitution for the full fibre, chronological averaging, and use of B^T as the forward homology cocycle. It also excludes an unproved direct sum or product over primes. No Riemann zeros or prime tables are read, trained on, or scored.